

Smart Traffic Management at busy Traffic light Junctions across India using Internet of things and Artificial Intelligence.

Ujjwal Dwivedi

¹Student, Department of Information Technology,
Dr. Akhilesh Das Gupta Institute of Technology and
Management.
New Delhi, Delhi 110053, India.
e-mail: dwivediujjwal@gmail.com

Vansh Gupta

¹Student, Department of Electronics and Communication
Engineering,
Dr. Akhilesh Das Gupta Institute of Technology and
Management.
New Delhi, Delhi 110053, India.
e-mail: vgvansh25@gmail.com

Abstract - This paper deals with the idea of using the concepts like internet of things and artificial intelligence in traffic signal control for better and more efficient traffic management at busy traffic light junctions in all big cities across India. The paper tries to demonstrate the audience how use of Artificial intelligence and internet of things in traffic lights at junctions can be used to prevent inefficient flow of traffic through a junction. We can record the flow and density of traffic coming from each road constituting the junction at different times of the day (i.e. regular hours and peak hours) and can develop a system that will analyze this data in real time. All The traffic signals at a junction can then be connected to this system which will link them together forming a network of traffic lights sharing data, which will enable them to work in sync with each other. The sequence of operation and duration of each traffic light at the junction will be controlled by the system to ensure a consistent flow of traffic in all directions through the junction depending upon the real time traffic density. With the help of Artificial intelligence the system can learn and modify the timing and operation sequence to maximize efficiency. The use of this system will help in solving problems like slow moving traffic, unnecessary road congestion, vehicular pollution, delays and increased travel time. It will help improve the existing traffic conditions and enhance the overall travel experience.

Keywords - Artificial intelligence, Internet of things, Traffic congestion, Traffic signal control.

I. Introduction

With the rise in population, the demand for transportation is also increasing rapidly every year. Due to the increasing number of vehicles and overuse of roads traffic jam is becoming a major issue in all the major cities across India.

Traffic Jam has a tremendous impact on the life of people. Vehicles have to wait for a long time to move in and out of the jam. It is one of the most serious problems in big cities that people have to deal with in daily life. Often it is caused due to narrow roads, poor traffic management and increased number of vehicles. India has the second largest road network in the world covering about 5.4 million kilometers. Major cities like

Delhi, Bangalore, Mumbai and many other cities of India are facing the issue of traffic congestion. The standard speed of many vehicles on the road of Indian cities is less than 20 kilometers per hour, and a journey of 10 kilometers could take 30 minutes or more. These traffic congestions cause a huge wastage of time and energy. The big cities of India are losing up to \$22 billion every year to traffic congestion, and this burden is carried by the commuters of these cities.



Fig 1: Traffic congestion at a busy Junction

India's road network has grown at an annual rate of 4% since 1951 but the number of vehicles has increased nearly 11%, choking roads and increasing pollution. In order to make traffic flow as efficient as possible and to make the most of our present road network we need to upgrade our traffic control system especially at traffic junctions. With the use of emerging technologies like 'Internet of things' and 'artificial intelligence' our traffic systems can be improved to cope with the increasing number of vehicles and to manage traffic better and more efficiently.

1.1. Internet of things

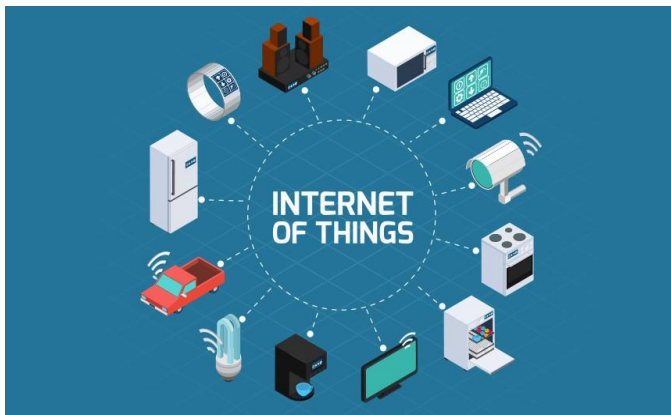


Fig 3: Internet of Things

The Internet of Things is the network of physical objects like devices, vehicles, buildings and other items—embedded with electronics, softwares, sensors, and network connectivity that enables these objects to collect and exchange data. In simple words, the IoT allows objects to be sensed and controlled remotely across existing network infrastructure, creating opportunities for more direct integration of the physical world into computer-based systems.

1.2. Artificial intelligence

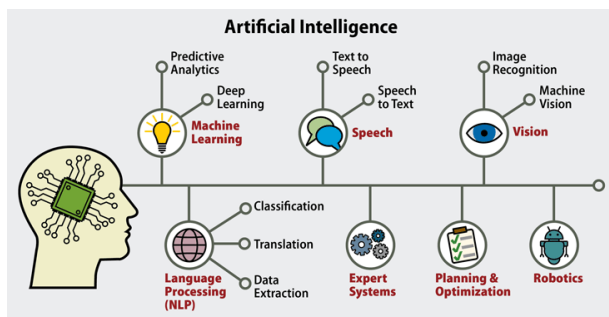


Fig 2: Diagram Explaining Artificial Intelligence

Artificial intelligence is the phenomenon which refers to the ability of a machine to interpret and learn from external data. The machine can make different decisions based on this data on its own and can learn to recognize a pattern that is perceives in its environment and takes actions that maximizes its chance of successfully achieving its goal. Artificial intelligence can be used to recognize objects with the help of cameras and image recognition software.

II. Existing Methods

Many different ideas and methods have been proposed for monitoring and controlling traffic over the period of time. In [1] the survey paper presents the various urban traffic control strategies which have been proposed or applied to prevent and treat congestion, Some basic definitions about congestion, such as primary and

secondary congestion, are remembered, Freeway control during congested situations is examined; the access control and exits control problems are developed. Approaches for intersection control are described: delay minimization, output flow rates maximization, LQ problem, congestion level feedback and democratic control.

Going forward towards a novel approach, it has been proposed to use measurements from a single station to detect anomalies in tragic flow. It was found that Anomalies are ascertained from deviations from the expected norms of tragic patterns calibrated at each individual station. This [2] paper discusses the development of a new novel time indexed anomaly detection algorithm. The author established norms as a time dependent function for each station by integrating the past “normal” tragic patterns for a given time period.

In the changing times keeping into consideration the existing traffic, a viable solution has been laid into place which will not only help the tracker to take the active control of the congestion but at the same time doing it in the most efficient manner. This paper [3] describes implementation of a real-time Visual tracking system equipped with an active camera. The system is intended for indoor human motion tracking. Real-time tracking is achieved using simple and fast motion detection procedures based on frame differencing and camera motion compensation. Results of on-line person tracking are presented.

In the contemporary times a more mechanized method has been introduced to channelize the control of traffic. In this paper [4] Background subtraction is a common computer vision task. The author analyzes the usual pixel-level approach. The author also develops an efficient adaptive algorithm using Gaussian mixture probability density. Recursive equations are used to constantly update the parameters but also to simultaneously select the appropriate number of components for each pixel.

In [5] a logical alternative was set to be proposed at the outset which would instill a new system for wireless traffic light controller, it makes a traffic policeman easily and effective control a road junction with a wireless remote control. The propose system help the policeman to control the junction by himself and be able to change conditions of the traffic flow dynamically.

However, over a significant time period it was into consideration present an intelligent Traffic Congestion Monitoring & Measurement System called Traffic Monitor to monitor and measure the road traffic congestions using probe vehicle. The concept of probe vehicle has come up in recent times for collecting real time traffic data. The system [6] provides an easy platform to analyze the traffic movement and congestion pattern.

In this paper [7] the author proposes an accurate and fast foreground detection technique for object tracking in videos with quasi-stationary backgrounds. The background is

modeled using a novel real-time kernel density estimations approach based on online histogram learning.

In the times of dynamism a possible alternative has come into place which proposes to suggest an intelligent traffic system. In this paper [8] a model is also proposed capable of managing intelligent traffic system using CCTV cameras and WAN. The proposed model will make the traffic signaling dynamic and automatic as well. Besides this, it will generate the dynamic messages for the users on the message boards to avoid congestion, reduce waiting time, pollution control, accident control and vehicle tracking.

Paving the way for further well planned method the author [10] proposed the algorithm to determine the number of vehicles on the road. The density counting algorithm works by comparing the real time frame of live video by the reference image and by searching vehicles only in the region of interest (i.e. road area). The computed vehicle density can be compared with other direction of the traffic in order to control the traffic signal smartly.

Further computation was taken into place and the author [11] devised a method based on optical flow calculation and Delaunay triangulation. Given a sequence of frames, firstly, the author extracted the corner feature points using ORB algorithm and compute optical flow vectors at the extracted feature points. Secondly, the author separates the optical flow vectors using K-Means clustering method. Third, we classify each cluster into camera and object motion using its mean scatter value. Finally, we represent the moving object using Delaunay triangulation.

The idea of presenting a well planned application for traffic management has been suggested by the author [12] when he investigates the prototype development of Internet of Things (IoT) application for traffic management.

The retrofitted traffic light control solution is proposed and developed to support decision making of police officers. The system can figure out the congestion level of each road at a specific intersection with the help of RFID technology.

To set the basics right, and devise a method which has a better tenacity the author has proposed this research. The paper [16] presents novel intelligent traffic administration system, based on Internet of Things, which is featured by low cost, high scalability, high compatibility, easy to upgrade, to replace traditional traffic management system and the proposed system can improve road traffic tremendously.

To cater to the emerging and urgent need of the mushrooming population about the traffic systems, the author has proposed this paper. The paper [18] analyses the ever growing urban population around the globe and discusses about the traffic systems in densely populated cities like Los Angeles and Amsterdam. Further, an advanced traffic management system is proposed, implemented using Internet of Things (IoT). The system is supported by a circuit embedded in the vehicle, which operates using RFID with clustered systems.

III. Proposed idea for developing an AI & IOT based traffic lights system to manage traffic flow at busy junctions in major cities of India

Traffic congestion at junctions is caused due to many factors such as improper junction design, narrow roads, heavy traffic, etc. but all these factors are circumstantial and are not present at every junction. The one factor that is present at every junction with a congestion problem is poor traffic management due to inefficient traffic signal control. This problem arises because the current traffic signal lights are preprogrammed and works on a fixed timer. The traffic signals at junctions are also not linked together and operate independently. They don't account for the fluctuations in traffic density taking place on all the roads in real time. For example, the light turns red for 30 seconds and then green for 30 seconds irrespective of the number of vehicles present at the signal and without taking into consideration the traffic flow on other roads connected to the junction and how it will affect it. Due to this congestions and delays take place at the junction.

In order to prevent traffic congestions at junctions and to make sure a smooth and efficient flow of vehicles through a junction, each road connected to the junction should have a traffic light at least half a kilometer before the junction's main traffic lights as shown in **Fig 4**, let's call these lights 'secondary lights' and the traffic lights present at the junction 'primary lights' for easier understanding. Secondary lights will be used to control the number of cars approaching the junction from every road at a given time just like a control valve and the primary lights will allow the cars to move through the junction. This way all the traffic on the roads will not pile up on the junction at once thereby causing congestion.

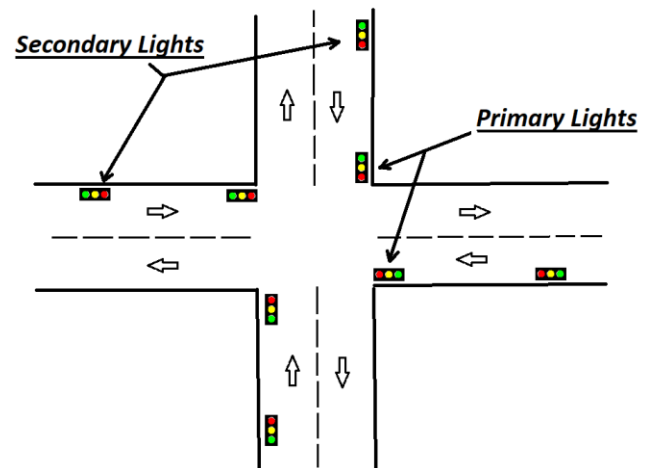


Fig 4: Concept of Primary & Secondary Lights

All the primary and secondary traffic lights will be connected to a central server forming a network. Cameras placed near the primary and secondary lights can be used to keep a track of number of vehicles present using real time

image recognition. This data will be uploaded to the central server controlling the junction traffic light network. Based on the traffic movement and density at the primary lights, the secondary lights will allow traffic movement towards the junction. The primary lights turn green allowing the traffic to move through the junction. Once the traffic clears the secondary lights will allow a controlled number of vehicles to move towards the junction into the controlled traffic area as Shown in Fig 5.

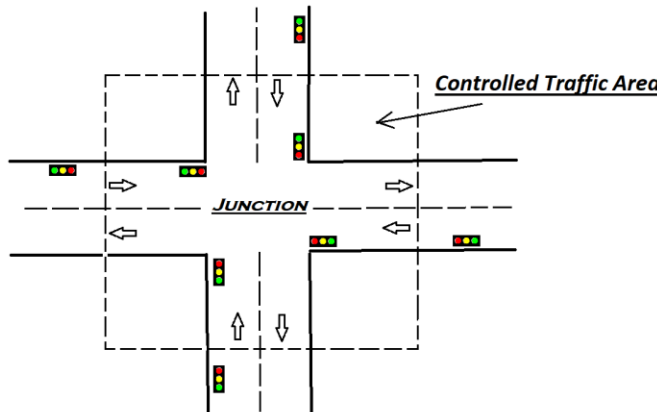


Fig 5: Traffic Controlled Area

This technique will be applied on all the roads connected to a junction. This way only that many vehicles will be present at the junction at a particular time that can smoothly move through without any unnecessary delay or congestion. Hence even when the traffic density will be high, cars will keep moving through the junction without causing congestion. The data collected at every light will be processed in real time. The road with the maximum number of vehicles in waiting will be given a higher priority and all the other traffic light timings will be adjusted accordingly. The system will automatically adjust and adapt based on the traffic density on each road at different times of the day like during rush hours in morning and evening or during festive season when the traffic is more than usual. This will ensure an overall smooth and the most efficient traffic movement possible through the junction at any particular time.

IV. Benefits of implementing this system

Traffic jam is a very serious problem which contributes to other major issues we are facing today like pollution, wastage of fuel and energy etcetera. Since most of the people have to deal with it on a daily basis they get psychologically affected and it creates a negative impact on the mind of a person. It also negatively affects work, education and has degrading effect on personal lives of people and finally to the progress of our country. Traffic jams results in non-productive activity and time wastage. People experience delays for their important work. This results in personal as well as professional losses. The traffic congestion and constant blowing of horns create excessive noise pollution. It increases stress and frustration among motorists and passengers which leads to unsafe driving, road mishaps and injuries. Businesses nowadays provide home delivery services. Such time bound businesses are strongly affected by the traffic jam. Emergency vehicles

like fire brigade and ambulance get stuck up in traffic jams that cause a delay in reaching the location thereby endangering lives. With the implementation of our proposed system, traffic jams at junctions can be considerably decreased and in many cases can be totally avoided which will prevent all the above mentioned problems that are a by product of these traffic jams and congestions.

V. Conclusion

Traffic jam is a serious issue in every big city of India that causes several problems for common people. It is important for the government to execute proper planning and come up with new and effective strategies like ours which will help in reducing the issue of traffic congestion. Our idea can help in enhancing the traffic management system which will make the execution of traffic related tasks much easier resulting in a smooth and efficient flow of traffic through busy junctions. Moreover, the government needs to utilize this latest and innovative technique in order to reduce the adverse effects of traffic jam on the roads. It consumes so much of time and energy unnecessarily and hence the loss of the nation.

VI. Acknowledgement

We would like to express our gratitude to our supervisors, Ankit Saxena and Dr. Pradeep Rohilla, who guided us throughout this project. We would also like to extend our gratitude to Mrs. Sharda Sharma, Executive Housekeeper at Radisson Blu Plaza Delhi Airport, for sharing her knowledge and experience with us. We would also like to thank our friends and family who supported us and offered deep insight into the study. Finally, our thanks go to all the people who have supported us to complete the research work directly or indirectly.

References

- [1] J. J. Henry and J. L. Farges, "TRAFFIC CONGESTION CONTROL," *IFAC Proceedings Volume 23*, pp. 177-182, 1989.
- [2] R weil, J WOOTTON, and A. GARCIA-ORTIZ, "Traffic Incident Detection," *Mathematical and Computer Modelling.*, vol. 27. 257-291. 10.1016/S0895-7177(98)00064-8. , 1998.
- [3] S. Ribarić, G. Adrinek, and S. Šegvić, "Real-Time Active Visual Tracking System," *Proceedings of the Mediterranean Electrotechnical Conference*, vol. - MELECON. 1. 231 - 234 Vol.1. 10.1109/MELCON.2004.1346816., June 2004.
- [4] Zoran Zivkovic, "Improved Adaptive Gaussian Mixture Model for Background Subtraction," *Proceedings of the 17th International Conference on Pattern Recognition*, SEPTEMBER 2004.
- [5] K Thatsanavipas, N. Ponganunchoke, and S. Mitatha, "Wireless Traffic Light Controller," *Procedia Engineering*, vol. 8. 190-194.

- 10.1016/j.proeng.2011.03.035., DECEMBER 2010.
- [6] Koushik Mandal et al., "Road Traffic Congestion Monitoring and measurement using active rfid and gsm technology," *2011 14th International IEEE Conference on Intelligent Transportation Systems (ITSC)*, april 2011.
 - [7] Alireza Tavakkoli and Kolawole Akintola, ""Robust Foreground Detection in Videos Using Adaptive Color Histogram Thresholding and Shadow Removal," *Proceedings of ISVC*, vol. vol. 2. 496-505. 10.1007/978-3-642-24031-7_50. , september 2011.
 - [8] Pratishtha Gupta, G.N Purohit, and Amrita Dadhich, "Approaches for Intelligent Traffic System : A Survey," *Int. J. Comput. Sci. Eng. 4.*, september 2012.
 - [9] Shruthi K R and Vinodha K, "PRIORITY BASED TRAFFIC LIGHTS CONTROLLER USING WIRELESS SENSOR NETWORKS," *International Journal of Electronics Signals and Systems (IJESS)*, vol. Vol-1, no. Iss-4, pp. ISSN: 2231- 5969, 2012.
 - [10] Naeem Abbas, Muhammad Tayyab, and Muhammad Tahir Qadri, "Real Time Traffic Density Count Using Image Processing," *International Journal of Computer Applications*, pp. 83. 16-19. 10.5120/14476-2736., december 2013.
 - [11] Sergey Starkov and Maksim Lukyanchenko, "Moving Object Detection in Video Streams Received from a Moving Camera," *CEUR Workshop Proceedings*, no. 1197, pp. 59-63, 2014.
 - [12] N Choosri, Y Park, S Grudpan, P Chuarjedton, and A Ongvisesphaiboon, "IoT-RFID Testbed for Supporting Traffic Light Control," *International Journal of Information and Electronics Engineering*, vol. 10.7763/IJEE.2015.V5.511., pp. 102-106, march 2015.
 - [13] Harsha H. Sanap and P. A. Ghonge, "Traffic Analysis Based On Image Processing," *International Journal of Science and Research (IJSR)*, no. ISSN (Online): 2319-7064, march 2015.
 - [14] Nidhi Daxini, Sachin Sharma, and Rahul Patel, "Real Time Animal Detection System using HAAR Like Feature," *International Journal of Innovative Research in Computer and Communication Engineering*, vol. 10.15680/ijrcce.2015.0306032., pp. 5177-5182, july 2015.
 - [15] I Made Oka Widyantara and Nyoman Putra Sastra, "Internet of Things for Intelligent Traffic Monitoring System:A Case Study in Denpasar," *International Journal of Computer Trends and Technology.*, vol. 10.14445/22312803/IJCTT-V30P130., pp. 169-173., december 2015.
 - [16] Hasan Omar Al-Sakran, "Intelligent Traffic Information System Based on Integration of Internet of Things and Agent Technology," (*IJACSA*) *International Journal of Advanced Computer Science and Applications*, vol. Vol. 6, No. 2, 2015.
 - [17] Khaled Abdul Rahman Jomaa, "An Artificial Intelligence Techniques And Simulation Model To Control A Traffic Jam System In Malaysia (Review Paper)," *Asian Journal of Business and Management* ISSN: 2321 – 2802, vol. Volume 04, no. Issue 01, february 2016.
 - [18] Mahesh Lakshminarasimhan, "IoT Based Traffic Management System," *research gate*, march 2016.
 - [19] Bharath Kumar Perumalla and M. Sunil Babu, "An Intelligent Traffic and Vehicle Monitoring System using Internet of Things Architecture," *International Journal of Science and Research (IJSR)ISSN (Online): 2319-7064*, november 2016.
 - [20] V. Varsani and S. Ramalingam, "VEHICLE DETECTION FOR TRAFFIC FLOW ANALYSIS," *2016 IEEE International Carnahan Conference on Security Technology (ICCST), Orlando, FL*, pp. pp. 1-8, january 2017.
 - [21] Tania Chawla and Dr Asmita A. Moghe, "Real-Time Traffic Detection using Horizontal and Vertical Scanning," *International Journal of Advanced Research in Computer Science and Software Engineering.*, no. 10.23956/ijarcsse/V7I6/0., pp. 798-804, june 2017.
 - [22] Divya vani. P., Aruna K, and Raghavendra Rao K, "Internet of Things-A Practical Approach to Certain Cloud Services using CC3200," *International Journal of Pure and Applied Mathematics*, vol. Volume 117 No. 10, pp. 125-128, october 2017.
 - [23] Ashok. P.V, SivaSankari S., Vignesh Mani, and Suresh Sankaranarayanan, "IoT Based Traffic Signalling System," *International Journal of Applied Engineering Research ISSN 0973-4562*, vol. Volume 12, Number 19, pp. pp. 8264-8269, 2017.
 - [24] Pampa Sadhukhan and Firoj Gazi, "An IoT based Intelligent Traffic Congestion Control System for Road Crossings," *2018 International Conference on Communication, Computing & Internet of Things (IC3IoT 2018), At S.R.M Engineering College, Chennai, India*, february 2018.
 - [25] Senthil Kumar Janahan et al., "IoT based smart traffic signal monitoring system using vehicles counts," *International Journal of Engineering & Technology.*, no. 7. 309. 10.14419/ijet.v7i2.21.12388., april 2018.
 - [26] Rashmi P. Nimkar and Chandrashekhar N. Deshmukh, "TRAFFIC DENSITY MONITORING AND CATTLE MENACE ALERT SYSTEM USING IOT," *IJRET: International Journal of Research in Engineering and Technology*, august 2018.
 - [27] Varsha Srinivasan, Yazhini Priyadharshini Rajesh, S Yuvaraj, and M Manigandan, "Smart traffic control with ambulance detection," *IOP Conference Series: Materials Science and Engineering.*, vol. 402. 012015. 10.1088/1757-899X/402/1/012015., september 2018.
 - [28] Chandana K K, Dr. S. Meenakshi Sundaram, Cyana D'sa, Meghana N Swamy, and Navya k, "A Smart Traffic Management System for Congestion Control and Warnings Using Internet of Things (IoT)," *Saudi Journal of Engineering and Technology*, may 2019.
 - [29] Simanta Shekhar Sarmah, "Artificial Intelligence in Automation," *research gate*, june 2019.

- [30] D.V.B Pragna, D.Laxma Reddy, and SVS Prasad, "IoT Driven Automated Object Detection Algorithm for Urban Surveillance System in Smart City," *International Journal of Engineering and Advanced Technology (IJEAT)* ISSN: 2249 – 8958, vol. Volume-8, no. Issue-6S3, september 2019.
- [31] Donato Impedovo, Fabrizio Balducci, Vincenzo Dentamaro, and Giuseppe Pirlo, "Vehicular Traffic Congestion Classification by Visual Features and Deep Learning Approaches:A Comparison," *Sensors*. 19. 5213. 10.3390/s19235213., november 2019.
- [32] HUA WEI, GUANJIE ZHENG, VIKASH GAYAH, and ZHENHUI LI, "A Survey on Traffic Signal Control Methods," *Association for Computing Machinery*, vol. Vol. 1, No. 1, Article, january 2020.
- [33] Mohan Raj A, Gayathri K, Navidharshini G, Renita Praisys S, and Srimathi W, "DENSITY BASED TRAFFIC CONTROL USING IoT," *International Journal of Computer Science and Mobile Computing*, vol. Vol.9, no. Issue.1, pp. pg. 132-136, january 2020.